

EXPERIMENT 6

UML CLASS DIAGRAM

Aim

To develop the UML Class Diagram for the Stock Maintenance System.

Objective

- To identify the core classes, attributes, and methods of the system.
- To establish the static structure and object-oriented design of the system.
- To define the relationships (associations, dependencies, and multiplicities) between different system components.

Tools Used

- Draw.io
- Lucidchart
- StarUML
- Creately
- Edraw Max

Theory

A UML Class Diagram is a static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among objects. It acts as the foundational blueprint for developing the system's code.

Key components include:

- **Classes:** Represented by boxes divided into three compartments (Class Name, Attributes, and Methods).
- **Visibility:** Defines access levels for data security (e.g., public +, private -).
- **Relationships:** Shows how classes interact, including associations (general communication), generalizations (inheritance), and dependencies.

System Classes

Based on the system requirements, the following primary classes were identified:

- **User:** The base data entity storing user credentials (id, name, username, password) and their designated role.

- StockManagementUI: The interface class that displays the dashboard, prompts logins, and captures user inputs for stock operations.
- UserAuthenticator: Handles the validation of login credentials and processes new user registrations.
- ProductController & AdminController: Manage the business logic for adding/updating products and managing system users.
- StockService: Executes the core inventory logic, such as checking stock status and adjusting quantities.
- Product: The data entity representing inventory items, storing attributes like quantity and supplier details.
- TransactionRecorder & ReportGenerator: Handle the logging of new transactions and the fetching of historical data for system reports.
- Transaction: The data entity storing records of stock movements, including timestamps, related product IDs, and transaction types.

Explanation

The Class Diagram illustrates a layered architecture. The interface layer (StockManagementUI) delegates user actions to specific controllers and services (ProductController, StockService, UserAuthenticator). These service classes contain the logic to manipulate the core data entities (Product, Transaction, User). Additionally, enumerations (RoleType, TransactionType) are linked to the entities to strictly define user permissions and transaction categories (Purchase vs. Sale).

Result

The UML Class Diagram for the Stock Maintenance System was successfully developed.

STOCK MANAGEMENT SYSTEM - UML CLASS DIAGRAM

